Linear Combinations, Span and Basis

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Vectors and feature vectors.
- **Leads to:** Linear transformations and matrices; Normal equation.
- **Compare with:** Multicollinearity.

1. Overview

Key point: Scaling vectors and adding them gives a **linear combination** (G-1091); all the vectors we can reach this way form the span; a set of vectors that reaches everything with no spare vector is a basis.

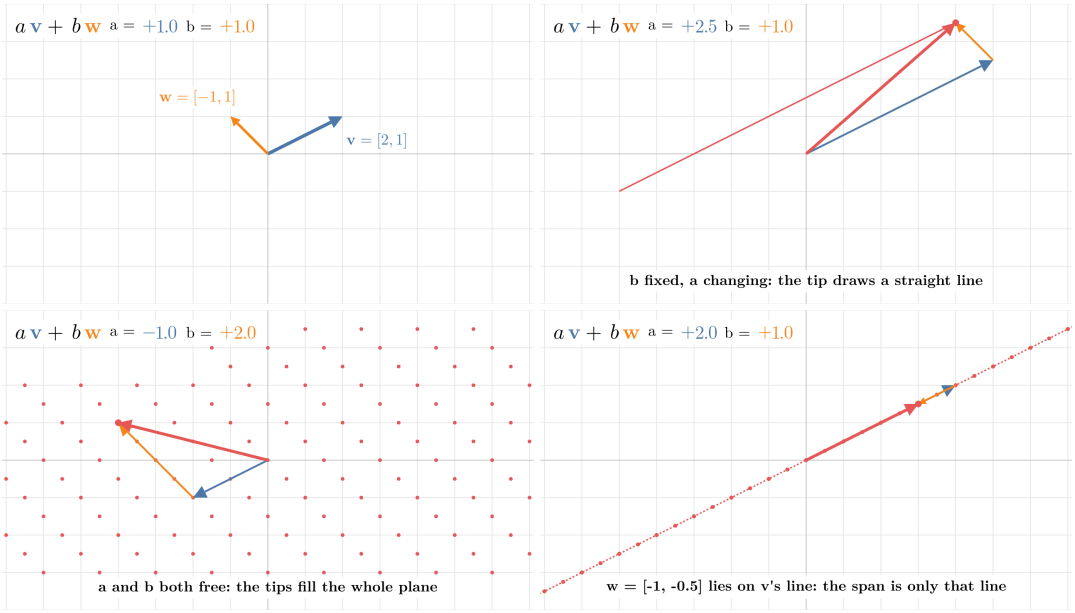


Figure 1: Linear combinations $a\mathbf{v} + b\mathbf{w}$: one free scalar traces a line, two free scalars fill the plane, and two vectors on one line only ever reach that line

Figure 1 shows the whole Note in motion. We take two vectors, scale each by a number, and add the results. Turning one number draws a line; turning both reaches every point of the plane, unless the two vectors lie on the same line.

What a vector is, and how a row of data becomes a feature vector, is taught in three views of a vector and **feature vectors** (G-771). **Scaling** (G-1746) a vector is in multiplying or dividing by a scalar. This Note adds the second operation, adding two vectors, and builds everything else from the two.

Extra: The geometric pictures in this Note and the next four (linear transformations, matrix multiplication, duality, eigenvectors) follow Sanderson's *Essence of Linear Algebra* (3Blue1Brown), the resource the linear algebra roadmap recommends.

2. Three ways to see a vector

Key point: A vector is an arrow (physics), a list of numbers (computer science), or anything that can be added and scaled (mathematics); linear algebra is the translation between the first two.

There are three related views of what a vector is (recapped from three views of a vector):

- **The arrow view (physics).** A vector is an arrow with a length and a direction. In linear algebra we almost always draw it with its tail at the origin (the point $[0, 0]$ where the axes cross).
- **The list view (computer science).** A vector is an ordered list of numbers. A house described by its area and its price is the 2D vector [area, price]; the order matters. Each number is one **feature** (G-772) (an input variable, one column of the data table), and the list view is the **feature vector** (G-771) of ML.
- **The abstract view (mathematics).** A vector is anything for which "add two of them" and "multiply one by a number" make sense. This view says that those two operations are what all of linear algebra is built on.

The coordinates of a vector link the first two views. They are instructions for walking from the tail at the origin to the tip: the first number says how far to walk along the x-axis (right if positive, left if negative), the second how far parallel to the y-axis. Every list of numbers gives exactly one arrow, and every arrow gives exactly one list.

The power of linear algebra lies in moving between the views. For data, a table of numbers becomes a picture of points we can reason about. For computer graphics or physics, a picture of space becomes numbers a computer can crunch.

Extra: One collection of vectors, two pictures. A single vector is best drawn as an arrow. A large set of vectors, such as all 150 iris flowers (150 **observations** (G-1374): one observation is one record, a row of the data table), is best drawn as points: one point at the tip of each arrow. The points picture is why a scatter plot of a dataset is a picture of its feature vectors (Figure 2).

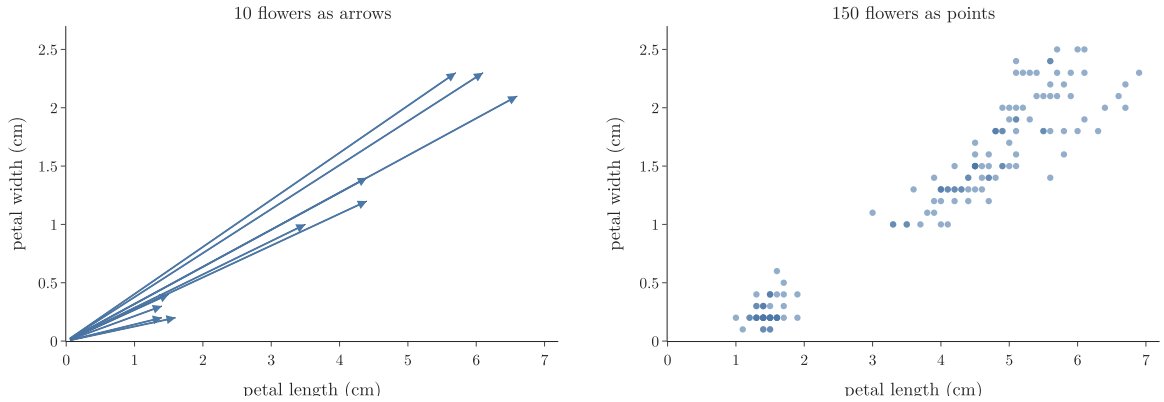


Figure 2: Iris flowers as feature vectors [petal length, petal width]: 10 of them as arrows from the origin (left), all 150 as points at the arrow tips (right).

3. Adding two vectors

> **Key point:** To add two vectors, put the tail of the second at the tip of the first; numerically, add matching components.

3.1 Tip to tail

> **Key point:** Each vector is a step; adding two vectors means taking one step and then the other.

Think of each vector as a movement: a step of some length in some direction. Taking the step $\mathbf{v}$ and then the step $\mathbf{w}$ lands us at the same place as one single step, their sum (**vector addition**, G-2080) $\mathbf{v} + \mathbf{w}$.

So to draw a sum, we slide the second arrow so that its tail sits on the tip of the first. The arrow from the origin to the new tip is the sum (Figure 3, left). Tip-to-tail drawing is the one place in linear algebra where we let a vector leave the origin. Tip-to-tail adding is the same idea as adding numbers on a number line: 2 steps right and then 5 steps right is 7 steps right.

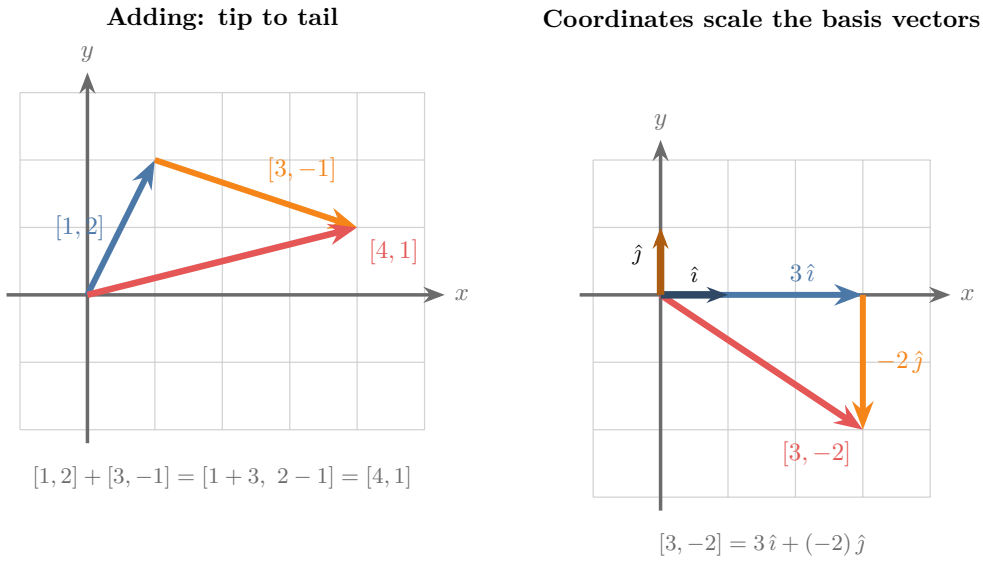


Figure 3: Left: adding $[1, 2]$ and $[3, -1]$ tip to tail. Right: the coordinates $[3, -2]$ as scaled basis vectors

3.2 Adding components

> **Key point:** Do all the horizontal steps, then all the vertical steps: the sum adds the x-components and the y-components separately.

1. **In words:** add the first components together, then the second components, and so on.
2. **Formula:** for two n-dimensional vectors,

$$\mathbf{v} + \mathbf{w} = [v_1 + w_1,\ v_2 + w_2,\ \ldots,\ v_n + w_n]$$

3. **Example:** walking $[1, 2]$ then $[3, -1]$ is 1 right, 2 up, 3 right, 1 down. Regrouping the horizontal and the vertical moves,

$$\begin{aligned} &[1, 2] + [3, -1] \\ &= [1 + 3,\ 2 + (-1)] \\ &= [4, 1] \end{aligned}$$

Both vectors must have the same number of components. With **scaling** (G-1746) (multiply every component by the number, which stretches, squishes or flips the arrow), we now have the two operations of linear algebra. A number used to scale a vector is called a **scalar** (G-1743), which is why "scalar" and "number" are used almost as synonyms.

> **Python:** NumPy adds vectors component by component.

```
import numpy as np

v = np.array([1, 2])
```

```
w = np.array([3, -1])
v + w        # array([4, 1])
2 * v        # array([2, 4]): scaling
```

4. Coordinates are scalars: basis vectors

> **Key point:** The coordinates $[3, -2]$ mean "3 times $\hat{\imath}$ plus -2 times $\hat{\jmath}$", where $\hat{\imath}$ and $\hat{\jmath}$ are the unit vectors along the axes.

The xy-plane has two special vectors:

- $\hat{\imath}$ ("i-hat"): length 1, pointing right along the x-axis, $\hat{\imath} = [1, 0]$.
- $\hat{\jmath}$ ("j-hat"): length 1, pointing up along the y-axis, $\hat{\jmath} = [0, 1]$.

Now read each coordinate as a **scalar** (G-1743). In $[3, -2]$, the 3 stretches $\hat{\imath}$ to 3 times its length, and the -2 flips $\hat{\jmath}$ and stretches it to twice its length. Adding the two scaled vectors gives the vector (Figure 3, right):

$$[3, -2] = 3\,\hat{\imath} + (-2)\,\hat{\jmath} = [3, 0] + [0, -2]$$

So every vector is a sum of scaled $\hat{\imath}$ and $\hat{\jmath}$. Together, $\hat{\imath}$ and $\hat{\jmath}$ are called the **standard basis** (G-1869) of the plane: the vectors that the coordinates scale.

In 3D a third unit vector $\hat{k} = [0, 0, 1]$ joins them, and in n dimensions there are n of them. A **feature vector** (G-771) reads the same way: the flower $[5.1, 3.5, 1.4, 0.2]$ is 5.1 units of "sepal length" plus 3.5 units of "sepal width", and so on, one unit vector per **feature** (G-772).

5. Linear combinations

> **Key point:** Scaling some vectors and adding the results is a linear combination; fixing all **scalars** (G-1743) but one makes the tip move along a straight line.

Scaling (G-1746) two vectors and adding them,

$$a\,\mathbf{v} + b\,\mathbf{w}$$

with a and b any numbers, is called a **linear combination** (G-1091) of $\mathbf{v}$ and $\mathbf{w}$. With more vectors, we pick one scalar per vector: $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k$.

One way to remember the word "linear": fix b and let only a change. The tip of $a\mathbf{v} + b\mathbf{w}$ then runs along a straight line, parallel to $\mathbf{v}$ (Figure 1, the stage "b fixed, a changing: the tip draws a straight line").

5.1 Other bases give other coordinates

> **Key point:** Any two vectors that do not line up can serve as a **basis** (G-262); the same arrow then gets different coordinates.

Nothing forces us to use $\hat{\imath}$ and $\hat{\jmath}$. Take $\mathbf{v} = [1, 1]$ and $\mathbf{w} = [1, -1]$ instead. Every 2D vector is still a **linear combination** (G-1091) of them, but with different scalars.

1. **In words:** find the **scalars** (G-1743) a and b that rebuild the vector from the new basis vectors; they are its coordinates in that basis.
2. **Formula:**
$$a\,\mathbf{v} + b\,\mathbf{w} = \mathbf{x}$$
3. **Example:** for $\mathbf{x} = [3, -2]$, the x-components and the y-components give two equations:
$$a + b = 3$$
$$a - b = -2$$
Adding the two equations:
$$2a = 1$$
$$a = 0.5$$
Putting a back into the first equation:
$$b = 3 - 0.5 = 2.5$$
Check:
$$0.5\,[1, 1] + 2.5\,[1, -1]$$
$$= [0.5 + 2.5,\ 0.5 - 2.5]$$
$$= [3, -2]$$

Figure 4 shows this example. Watch the orange arrow: it never moves. Only the grid we measure it with changes, from the square grid of $\hat{\imath}$ and $\hat{\jmath}$ to the tilted grid of $\mathbf{v}$ and $\mathbf{w}$, and with it the two numbers.

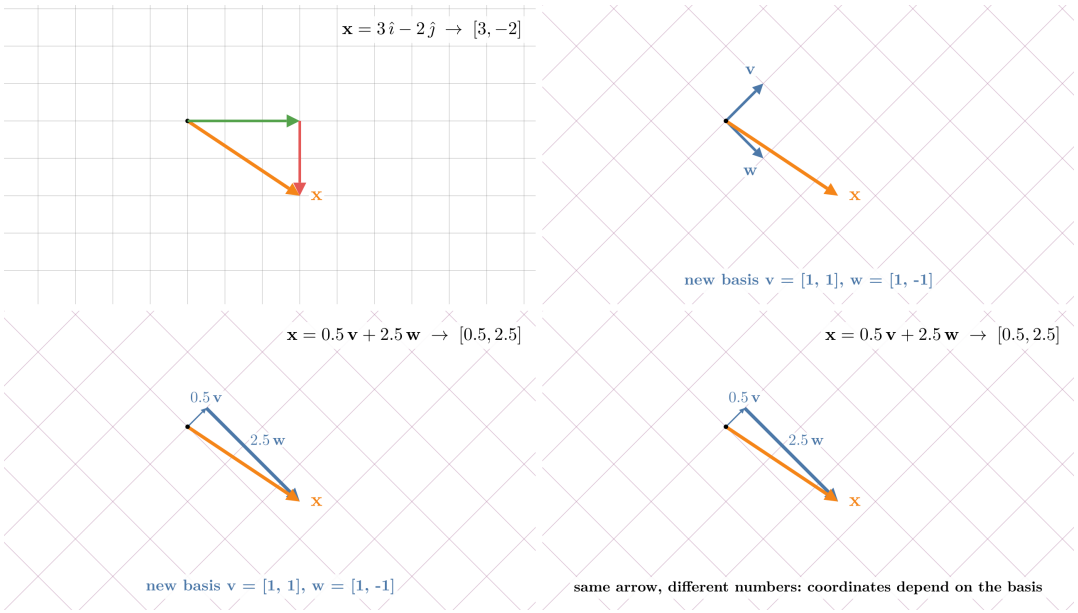


Figure 4: The arrow $[3, -2]$ measured with two bases: $3\,\hat{\imath} - 2\,\hat{\jmath}$ on the standard grid, and $0.5\,\mathbf{v} + 2.5\,\mathbf{w}$ on the grid of $\mathbf{v} = [1, 1]$ and $\mathbf{w} = [1, -1]$

So the same arrow is $[3, -2]$ in the **standard basis** (G-1869) and $[0.5, 2.5]$ in the basis $\mathbf{v}, \mathbf{w}$. Whenever we write a vector as numbers, we have silently chosen a basis.

> **Python:** Finding the coordinates in a new basis means solving a small system of equations (here $a + b = 3$ and $a - b = -2$). `np.linalg.solve` takes the basis vectors as the columns of a matrix.
>
> ```
> # v and w as the columns of B
> B = np.column_stack([[1, 1], [1, -1]])
> np.linalg.solve(B, [3, -2])      # array([0.5, 2.5])
> ```

> **Extra:** Changing the basis is everyday ML. PCA (principal component analysis, see PCA step by step) re-describes each data point in a new basis made of the principal components: the new coordinates are the PC1, PC2, ... values. Word embeddings and the hidden layers of a neural network can be read in the same spirit: each describes the data by new numbers that the model found useful, a learned **representation** (Bengio et al. 2013).

6. Span

> **Key point:** The span of some vectors is every vector we can reach with their **linear combinations** (G-1091): for two 2D vectors, the whole plane, or only a line if they line up.

The **span** (G-1838) of a set of vectors is the set of all their linear combinations. It answers one question: using only **scaling** (G-1746) and adding, where can we get to?

6.1 Two vectors in 2D

> **Key point:** Two vectors pointing in different directions **span** (G-1838) the whole plane; two vectors on one line span only that line.

Let both **scalars** (G-1743) in $a\mathbf{v} + b\mathbf{w}$ range freely. Three things can happen:

- **Most pairs:** the tips reach every point of the plane. In Figure 1 (the stage "a and b both free: the tips fill the whole plane"), $\mathbf{v} = [2, 1]$ and $\mathbf{w} = [-1, 1]$ build a slanted grid of reachable points, and filling in all the values between gives the whole plane.
- **Two vectors on one line:** if $\mathbf{w}$ is a multiple of $\mathbf{v}$, every combination lies on that line. In Figure 1 (the last stage, "w = [-1, -0.5] lies on v's line"), $\mathbf{w} = [-1, -0.5] = -0.5\,\mathbf{v}$, so the span is the line through the origin along $\mathbf{v}$.
- **Both vectors zero:** the span is the origin alone.

6.2 Vectors in 3D

> **Key point:** Two 3D vectors **span** (G-1838) a flat sheet through the origin; a third vector either lies on that sheet and adds nothing, or leaves it and unlocks all of 3D space.

In 3D, two vectors that do not line up span a plane through the origin. Picture two knobs, one per **scalar** (G-1743): as we turn them, the tip of $a\mathbf{v} + b\mathbf{w}$ sweeps out a flat sheet. In Figure 5, $\mathbf{v}_1 = [2, 0, 1]$ and $\mathbf{v}_2 = [0, 2, 1]$ span the blue plane, whose points are all $[2a, 2b, a + b]$.

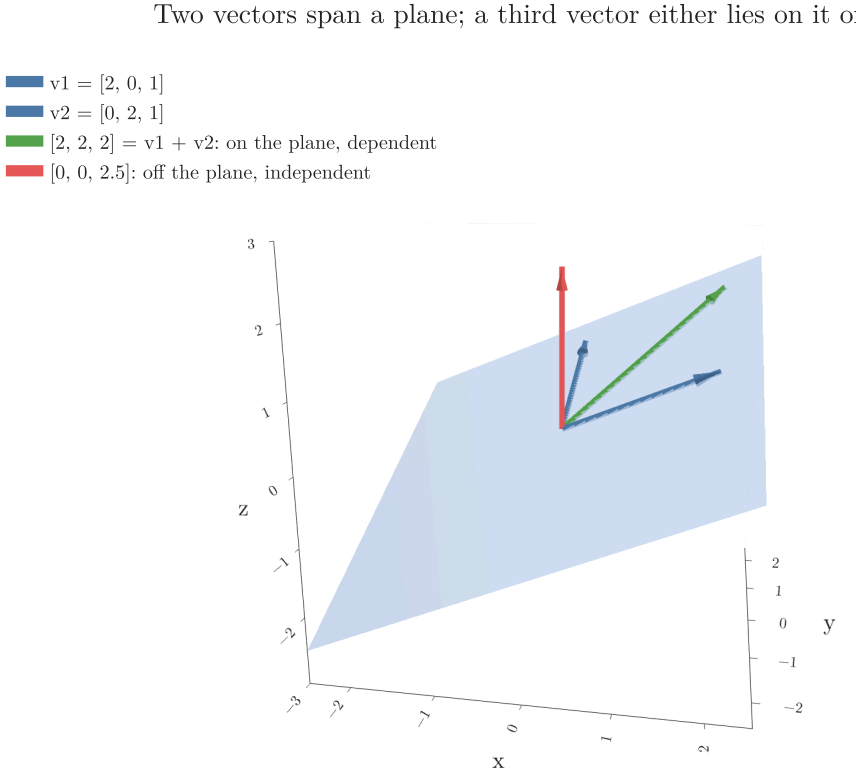


Figure 5: The plane spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$, a third vector on it (green) and one off it (red)

A third vector, with a third scalar, gives two cases:

- **On the plane** (green, $[2, 2, 2]$, the sum of $\mathbf{v}_1$ and $\mathbf{v}_2$): its scaled copies stay on the sheet. The span does not grow.
- **Off the plane** (red, $[0, 0, 2.5]$): **scaling** (G-1746) it lifts the sheet up and down, sweeping it through all of space. The span becomes the whole of 3D.

A third vector chosen at random almost always lands off the plane.

7. Linear dependence and independence

> **Key point:** Vectors are linearly dependent when one of them is a combination of the others, so it can be removed without shrinking the **span** (G-1838); otherwise they are independent.

When a vector adds nothing to the span (the green vector in Figure 5, or two 2D vectors on one line), we need a word for it. Vectors are **linearly dependent** (G-1101) when at least one of them can be written as a **linear combination** (G-1091) of the others: removing it does not shrink the span. They are **linearly independent** (G-1102) when each one adds a new direction, a new dimension, to the span.

Figure 6 animates the two cases of Figure 5. Watch the sheet: **scaling** (G-1746) the green vector only slides its tip along the sheet, so the span stays a plane; scaling the red vector lifts copies of the sheet up and down, and the stack of sheets fills all of 3D. The picture follows Sanderson's *Essence of Linear Algebra*, chapter 2 (3Blue1Brown).

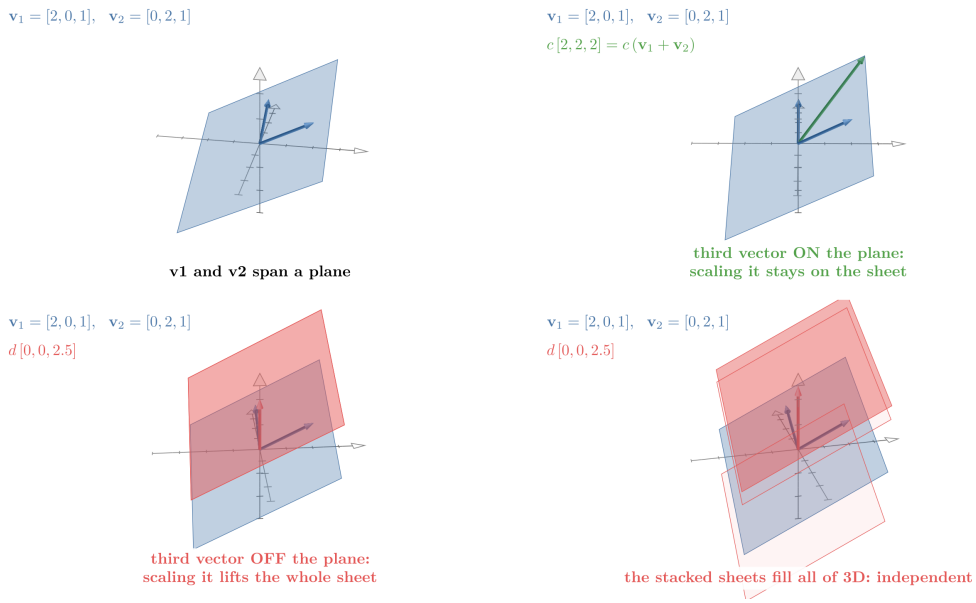


Figure 6: A third vector on the plane of $\mathbf{v}_1$ and $\mathbf{v}_2$ (green, dependent) only moves within the plane; a third vector off it (red, independent) sweeps the plane through all of space

With numbers:

- $[1, 2]$ and $[2, 4]$ are dependent, since $[2, 4] = 2\,[1, 2]$. Their span is a line.
- $[1, 2]$ and $[3, 1]$ are independent. Their span is the whole plane.
- $[2, 0, 1]$, $[0, 2, 1]$ and $[2, 2, 2]$ are dependent, since the third is the sum of the first two.
- $[1, 2]$, $[3, 1]$ and $[4, 3]$ are dependent, since $[4, 3] = [1, 2] + [3, 1]$. The first two already span the plane, so a third 2D vector has nowhere new to point.
- $[2, 0, 0]$, $[0, 3, 0]$ and $[0, 0, 5]$ are independent. Each one has a non-zero component where the other two have 0, so no combination of two of them can build the third. Their span is all of 3D.

Python: NumPy's `matrix_rank` counts how many independent directions a set of vectors has. Put the vectors in as columns; if the count is smaller than the number of vectors, they are dependent.

```
rank = np.linalg.matrix_rank
rank(np.column_stack([[1, 2], [2, 4]]))    # 1
rank(np.column_stack([[1, 2], [3, 1]]))    # 2
V3 = np.column_stack([[2, 0, 1], [0, 2, 1], [2, 2, 2]])
rank(V3)                                   # 2
```

The number is the **rank** (G-1627) of the matrix. This matrix rank is a different meaning from the rank of a tensor (its number of axes) in rank, axes, shape and size.

Extra: Linear dependence between the columns of a dataset is exactly the **multicollinearity** (inputs that depend on each other) of the dummy variable trap. The dummy columns of one category always add up to 1, the column of ones used for the intercept, so one of them is a linear combination of the others. The dependence is why one dummy column is dropped, and why the normal equation (the formula that solves least squares in one step) fails when columns are dependent (see the cost of the inverse).

Extra: A linear regression's predictions are a linear combination of the input columns:

$$\hat{y} = \beta_0 \cdot \mathbf{1} + \beta_1 \mathbf{x}_1 + \cdots + \beta_m \mathbf{x}_m$$

Here $\mathbf{1}$ is the column of ones and $\mathbf{x}_j$ are whole columns of n values. So every possible prediction vector lies in the span of the columns. Least squares fitting picks the point of that span closest to the true y: its prediction vector is the orthogonal projection of y onto the span of the columns (the closest point, found by dropping a perpendicular; see the picture: a projection) (ESL §3.2, Figure 3.2).

8. Basis

Key point: A basis of a space is a set of **linearly independent** (G-1102) vectors that spans it: enough vectors to reach everything, and none to spare.

Putting the words together: a **basis** (G-262) of a space is a set of linearly independent vectors whose **span** (G-1838) is the whole space. This matches Section 4: the basis vectors are what the coordinates scale.

- **"Spans the space":** every vector of the space is a **linear combination** (G-1091) of the basis, so every vector gets coordinates.
- **"Linearly independent":** no basis vector is redundant, so the coordinates are unique. With a spare vector, the same arrow could be written in more than one way.

$\hat{\imath}, \hat{\jmath}$ is a basis of the plane, and so is $[1, 1], [1, -1]$. But $[1, 2], [2, 4]$ is not: it spans only a line. Every basis of the plane has exactly 2 vectors, and every basis of n-dimensional space has exactly n. That count is what the **dimension** of a space means (MML §2.6.1).

Figure 7 shows the three cases with the target $[3, -2]$:

- **a basis**, $[1, 1]$ and $[1, -1]$: exactly one combination, $0.5\,\mathbf{v} + 2.5\,\mathbf{w}$;
- **not spanning**, $[1, 2]$ and $[2, 4]$: every combination stays on one line, so $[3, -2]$ is out of reach;
- **a spare vector**, $[1, 0]$, $[0, 1]$ and $[1, 1]$: many combinations work, for example $3, -2, 0$ and $2, -3, 1$, so the coordinates are not unique.

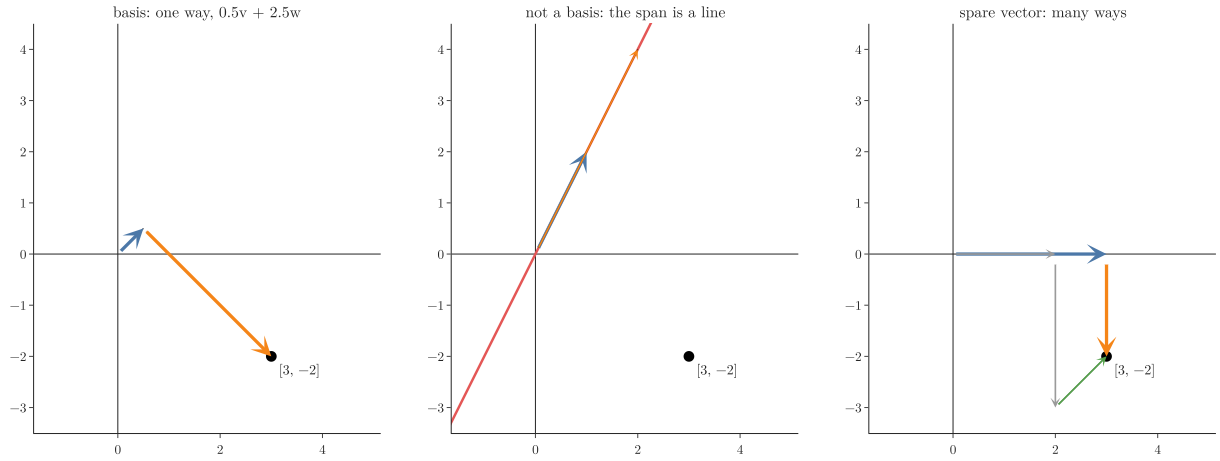


Figure 7: Reaching [3, −2]. Left: the basis [1, 1], [1, −1] gives one combination. Middle: [1, 2] and [2, 4] span only the red line. Right: with a spare vector, two different combinations (blue and orange, grey and green) both arrive.

9. Summary

Idea	What it is	Example	Why it matters
Vector addition	add matching components; tip to tail	$[1,2]+[3,-1]=[4,1]$	one of the two operations everything else is built on
Standard basis	unit vectors along the axes	$\hat{\imath}=[1,0]$, $\hat{\jmath}=[0,1]$	the vectors that ordinary coordinates scale
Coordinates	the scalars on the basis vectors	$[3,-2]=3\hat{\imath}-2\hat{\jmath}$	they depend on the basis; PCA gives data new coordinates in a new basis
Linear combination	scale, then add	$0.5\,[1,1]+$ $2.5\,[1,-1]=[3,-2]$	a linear regression prediction is one, built from the input columns
Span	everything reachable by linear combinations	$[2,1],[-1,1]$: the whole plane	says where scaling and adding can reach
Linearly dependent	one vector is a combination of the others	$[2,4]=2\,[1,2]$	it adds nothing to the span; dependent columns are multicollinearity
Basis	independent and spanning	$[1,1],[1,-1]$	every vector gets exactly one set of coordinates

- Linear algebra is built on two operations: adding vectors and scaling them, because **linear combinations** (G-1091), span and basis are all defined from these two.
- Coordinates depend on the chosen **basis** (G-262); the same arrow has different numbers in different bases, so writing a vector as numbers always means a basis was chosen.
- Two independent vectors **span** (G-1838) a plane, three span 3D space; a dependent vector adds nothing, because it is already a combination of the others.
- Dependent **features** (G-772) are multicollinearity, and they break the normal equation, so one dummy column is dropped in one-hot encoding.
- So **scaling** (G-1746) and adding decide which vectors we can reach (the span), and a basis reaches them all with no spare vector.

10. Sources

Built from

- Sanderson, G. (3Blue1Brown), "Vectors | Chapter 1, Essence of linear algebra", 2016, 3blue1brown.com/lessons/vectors, https://www.youtube.com/watch?v=fNk_zzaMoSs

- Sanderson, G. (3Blue1Brown), "Linear combinations, span, and basis vectors | Chapter 2, Essence of linear algebra", 2016, 3blue1brown.com/lessons/span, https://www.youtube.com/watch?v=k7RM-ot2NWY
- Khan Academy, "Introduction to linear independence | Vectors and spaces | Linear Algebra | Khan Academy", YouTube, https://www.youtube.com/watch?v=CrV1xCWdY-g

Other references

- Bengio, Y., Courville, A. and Vincent, P. (2013). "Representation Learning: A Review and New Perspectives". *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35(8).
- Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer. Section 3.2 (ESL).
- Deisenroth, M. P., Faisal, A. A. and Ong, C. S. (2020). *Mathematics for Machine Learning*. Cambridge University Press. Section 2.6.1 (MML).

11. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Linear combination (G-1091)	A sum of vectors each multiplied by a number, $a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k$; changing the numbers reaches different vectors, and all the vectors it can reach form the span.
Vector addition (G-2080)	Adding matching components; geometrically, placing the second arrow's tail at the first arrow's tip.
Standard basis ($\hat{\imath}$, $\hat{\jmath}$) (G-1869)	The unit vectors along the axes, $[1, 0]$ and $[0, 1]$ in 2D; every vector is a sum of scaled copies of them, and its coordinates are those scales.
Basis (G-262)	A set of vectors that are linearly independent and whose span is the whole space, so every vector can be built by scaling the basis vectors and adding them; a vector's coordinates are those scales.
Span (G-1838)	The set of all linear combinations of some vectors: every point reachable by scaling and adding them, such as the whole plane for $\hat{\imath}$ and $\hat{\jmath}$.

Term	Meaning
Linearly dependent (G-1101)	Said of a set of vectors in which at least one can be built from the others by scaling and adding (a linear combination), so it adds nothing to the span.
Linearly independent (G-1102)	Said of a set of vectors in which none can be built from the others, so each one adds a new direction to the span.
Feature vector (G-771)	The vector of input values of one data point.
Scaling (a vector) (G-1746)	Multiplying or dividing every component of a vector by the same scalar, such as $2 \times [1, 3] = [2, 6]$; it makes the arrow longer or shorter along the same line.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Scalar (G-1743)	A single number, such as 5 or -6: a 0D tensor; multiplying a vector by it scales the vector, which is where the name comes from.
Rank (of a matrix) (G-1627)	The number of linearly independent columns of a matrix, that is, how many directions its columns really span; a column that is a combination of others does not add to it.